CASPER STUDIOS · PM CHALLENGE · WRITTEN SUBMISSION

Where each returned item *should go.*

A short written companion to the deck. Discovery, PRD, evaluation plan, and one feature broken into engineering-ready tickets — all in one document, so you can skim it, mark it up, or send it around.

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01 · DISCOVERY

How returns disposition works today, the pain points, baseline metrics, and where AI delivers the step change.

02 · PRODUCT REQUIREMENTS

Problem, users, jobs-to-be-done, MVP scope, metrics, risks and the open questions discovery must answer.

03 · AI EVALUATION PLAN

Four lifecycle checkpoints and a fixed rhythm to keep the model honest after launch.

04 · TICKETS

The disposition-recommendation feature decomposed into six engineering-ready Linear-style tickets.

01 Discovery · Current-state assessment

~1 PAGE

Empty shelves make the news. Returns lose more money, and no one is watching. This section maps how returns disposition operates at an apparel distribution centre today, names the friction, sets baseline metrics a returns manager already tracks, and points at where AI delivers step-change improvement.

THE WORKFLOW, FIVE STEPS

01	02	03	04 · The call	05
CUSTOMER	WORKER	SYSTEM	WORKER	SYSTEM
Return started online or in store	**Receive & inspect (sight and touch)**	**Look up item · inventory basics only**	**Manual disposition · gut feel, seconds**	**Log the decision, update inventory**

Step 4 is one of six paths — **resell as new · outlet · refurbish · liquidate · donate · destroy** — chosen by a worker in under a minute, without the data to pick the best one.

BASELINE METRICS · WHAT A RETURNS MANAGER ALREADY TRACKS

52%
of returned apparel is **not** resold at full price
EIGHTX · REVERSE LOGISTICS

20–30%
of an item's original value eaten by the work of processing the return
EIGHTX

45s
average time a worker takes to decide where a single item should go
OPTORO REVERSE-OPS BENCHMARKS

$890B
US returns expected 2025 — 16.9% of sales, ~24.5% online; apparel leads
NRF 2024

WHERE THE FRICTION LIVES

MISSING INFORMATION
The worker cannot see what each channel would pay for this exact item, right now.

INCONSISTENT CALLS
Two workers can inspect the same item and route it differently. Same item, different result.

SPEED OVER QUALITY
When volume is high, workers pick the fastest option. That is almost always liquidate or markdown.

PEAK OVERLOAD
Holiday and season-change spikes flood the desk. Manual grading cannot keep up.

> **FIELD QUOTE** "By 2pm I stop reading tags. If it looks worn, it goes to liquidation. I'll take the hit on value over falling behind." — composite, apparel returns floor lead

WHERE AI DELIVERS STEP-CHANGE IMPROVEMENT

Traditional software gives the worker a checklist; rules engines route by SKU. Neither closes the gap between the fastest call and the most valuable one, because the right answer depends on live signals that change hourly. AI is the only approach that can do all three of these things at once:

01 · SCORE
All six channels in one second
Using signals that were fresh 15 minutes ago — channel price, capacity, refurb cost, demand — not last week's assumptions.

02 · EXPLAIN
Every call in plain English
"Outlet demand up 18%, refurb is expensive, main store full." The worker trusts what they can read, and overrides what they can't.

03 · LEARN
From every override, every shift
Each override is a labelled disagreement — the tightest feedback loop in reverse logistics, compounding week over week.

Optoro already proves the math: 10–30% recovery lift, 100M+ items routed at Fortune-500 scale (IKEA, Best Buy, Staples). The question is not *does this work* — it is where we run it and how we go further.

Product Requirements Document

PROBLEM STATEMENT

Apparel returns get routed by workers who cannot see what each channel would pay for this exact item, right now. Money leaks quietly, at scale, thousands of times a day, because the highest-value step in the entire reverse-logistics flow is made in seconds using gut feel and old rules. Fix that step and recovery rate — the number returns managers already care about — moves for the whole warehouse.

USERS & JOBS-TO-BE-DONE

WAREHOUSE WORKER
"Tell me where this item makes the most money, without slowing me down. I'll make the call."

RETURNS MANAGER
"Recover more value, stay on the right side of the rules, and keep the line moving during peak."

PLANNING & BUYING
"Show me returned inventory I can actually reuse, not something I find out about weeks late."

SOLUTION · A RECOMMENDATION, NOT A REPORT

The recommendation starts the moment the customer hits "return", not when the box arrives. Two phases:

PHASE 1 · WHEN THE RETURN IS STARTED ONLINE

First-guess recommendation, before the box ships

Uses the reason, the item, the customer's history, a photo if attached, and how the item is selling right now. Result: a starting state (e.g. *"Likely outlet · 71% confident · pending inspection"*) attached to the return record days before it arrives.

PHASE 2 · WHEN THE ITEM LANDS AT THE DC

Final call, sharpened with condition and live channel numbers

The worker scans, grades condition (A/B/C/D), and the same monitor shows a ranked list, plain-English reason, and one-tap confirm or override. Under 400ms end-to-end. The worker keeps the final call.

SCOPE OF VERSION ONE

IN V1

- One busy warehouse, one product category (apparel)
- First guess from the online return, sharpened at the DC
- Six channels: resell, outlet, refurbish, liquidate, donate, destroy
- Human confirms every call; overrides logged with a reason
- Analytics wired from day one — every decision + input snapshot lands in the warehouse within 24h

NOT IN V1

- Auto-action with no person in the loop
- Multiple warehouses or categories at once
- Condition grading from a photo alone (photos assist, do not decide)
- New hardware — same scanner, same monitor

METRICS · WHAT WE MOVE AND WHAT WE PROTECT

METRIC	DEFINITION	TARGET	ROLE
Recovery rate `= value recovered ÷ retail price`	How much of the original retail price we get back after a return, no matter which channel it ended up in.	**`+5 pts vs. baseline`** `90 days after launch`	NORTH STAR
% resold at full price	Items put back on the shelf and sold at full price within 30 days.	`+3 pts`	DRIVER
Override rate	How often the worker picks a different option than we suggest.	`≤ 25%, trending down`	TRUST DRIVER
Seconds per item	Time from scan to confirmed disposition.	`No slower than today`	GUARDRAIL
Items per hour, per worker	Throughput at the inspection station.	`Same or +5%`	GUARDRAIL

WHY % A $30 tee and a $650 Arc'teryx jacket both get "+5 pts of recovery", but the dollars back look very different. Recovery rate scales with item value, so the metric holds across categories and price points.

02 Product Requirements Document · continued

VALUE LOGIC

Sizing at 5M returned apparel items per year at ~$80 average retail, recovery-rate lift of +5 points is roughly **~$20M / yr** in extra recovered value. That's $1.50 on a $30 tee, $32.50 on a $650 jacket — the metric scales with item value automatically. This lands squarely inside Optoro's public 10–30% recovery-lift band; the +5 point target is the floor, not the ceiling.

KEY RISKS

R1 Data may not be there.
Live channel prices are the highest-value input and the hardest to guarantee.
MITIGATION · Fall back to last-known + confidence penalty; flag stale bundles.

R2 Workers have to trust it.
Trust is earned by knowing when to say “ask a person” and explaining every call in plain English.
MITIGATION · Confidence-gated UI, plain-English reasons, override always in one tap.

R3 New items lack history.
Cold-start SKUs have no signal for recovered-$ predictions.
MITIGATION · Category-level priors + human review queue until history builds.

R4 Integration is real work.
Real-time reads on warehouse + inventory systems are hard. Wrong first partner slows launch.
MITIGATION · Pick pilot on integration readiness first, volume second.

OPEN QUESTIONS · DISCOVERY ANSWERS THESE

Q1 Pricing signal quality
How do we price each channel accurately enough for the model to trust and learn from? Which upstream systems own that number, and how fresh do we need it?

Q2 Unusual items
Separate process, or safe default recommendation? Long-tail SKUs will exist forever; we need to decide whether they get a rules fallback or a “skip to human” path.

Q3 Photo-based condition grading in v1, or wait for v2?
Photos as an assist are cheap. Photos as the deciding signal are a different product — accuracy, false-positive tolerance, and regulatory implications all shift. My default is v2, but this is open.

Q4 Retrain trigger
At what override rate do we call a category stale and retrain? My starting threshold is 30% weekly override in a category, but the right number will come from the first month of live data.

NOT THE EASY PATH · THOUGHT-PROCESS NOTES

Three sexier alternatives I turned down, and why. Each of them shows the tradeoff v1 makes explicit — so future scope conversations start from a shared baseline.

PATH NOT TAKEN · 01

Condition grading from a photo

Sexier demo, worse product. Without live channel economics the recommendation is still uninformed. Condition grading is deferred; the channel-pricing service (DIS-102) is the highest-leverage backend piece to build first.

PATH NOT TAKEN · 02

Fully automated call, no human

A v3 and a compliance conversation. Every removed human-in-the-loop costs trust and traceability. Overrides are also our best training signal — cutting them off ends the learning loop.

PATH NOT TAKEN · 03

Multi-warehouse launch

Faster claim to scale, worse feedback signal. One busy DC gives us a clean A/B and enough volume to move the metric within 90 days. Integration risk drops from N-ways to one-way.

03 AI evaluation plan

An AI product decays without a watcher. This plan sits on top of the PRD metrics and defines the checkpoints across the lifecycle plus the rhythm to keep improving the model after launch. It answers three questions: is the model right, is it fair to the worker, and is it drifting.

FOUR LIFECYCLE CHECKPOINTS

BEFORE WE LAUNCH

Backtest on history

Replay 6 months of returns with known outcomes. The model must beat the human baseline on recovered $ before we ship anything to a live station.

AT LAUNCH

Live A/B

50/50 split across shifts. Recovery-rate lift is the primary metric; guardrails: no worse on seconds-per-item or throughput.

ONGOING

Watch every day

Override rate and prediction accuracy, per category and warehouse. Weekly review with the returns manager on the same dashboard.

SAFETY NET

Catch the bad calls

Alerts on destroy-vs-sellable, sustainability rule breaks, drift over threshold, and stale live-data bundles. Fires within 5 minutes with a link to the offending decision.

WHAT WE MEASURE — ONE TABLE

SIGNAL	HOW WE MEASURE IT	THRESHOLD / TARGET
Recovery-$ lift	Predicted vs. realized dollars, joined via the settlement back-fill. Primary A/B metric.	`+5 pts recovery rate at 90 days`
Prediction accuracy	MAE between predicted recovered $ and settled $, per category, per warehouse.	`≤ 15% MAE per category`
Override rate	Share of recommendations the worker changes; broken down by category and reason code.	`Global ≤ 25%, any category ≤ 30% weekly`
Safety events	Count of destroy-vs-sellable mismatches, sustainability rule breaks, stale-data bundles served.	`Zero destroy-vs-sellable per week`
Worker adoption	% of shifts running through the tool at each station.	`≥ 90% by day 45`
Latency	Time from scan to on-screen recommendation, p95 and p99.	`p95 ≤ 400ms · p99 ≤ 800ms`

HOW WE MEASURE IT · PLUMBING

- **Every decision persists with its input snapshot** (recommendation, confirmed action, override reason, condition, data bundle) — ticket DIS-105, non-negotiable from day one.
- **Realized $ back-fills from settlements** (outlet POS, refurb resale, liquidator invoice) — DIS-106. Recovery rate reflects settled, not intent.
- **Dashboards + alerts** filterable by warehouse and category — DIS-108. Safety-net alerts fire in 5 min; drift breach auto-triggers a shadow retrain.

RHYTHM · HOW WE MAKE IT BETTER

WEEKLY

Review overrides

Any category over 30% overrides in a week hits the retrain shortlist. Reviewed with the returns manager.

EVERY TWO WEEKS

Prune features

Drop signals that stopped moving the call. Add new ones that do. Keep the model lean and legible.

MONTHLY

Label the hard ones

Low-confidence items go to a human review queue. Their labels become the next training set.

QUARTERLY

Retrain & shadow-test

Retrain on last 90 days. Ship only if the candidate clearly beats the champion on the offline gates.

> **HIGHEST-VALUE SIGNAL** Overrides are our best training data — every "no" comes with a reason, and that closes the loop between what the model thinks and what actually works on the floor. Everything about the UI, the reason picker, and the persistence schema is designed to protect that signal.

04 Tickets · one feature decomposed

LINEAR-STYLE · ENGINEERING-READY

Feature: AI disposition recommendation at the inspection station (v1). The six tickets below cover the complete loop — from the first-guess call when a customer starts a return online, through the worker's confirm-or-override at the warehouse, to the decision log that feeds the next retrain. They ship as one release so v1 launches with its learning loop already wired. ★ marks the MVP core; priority, story points, and tags sit on the right of each ticket.

DIS-100 First-guess recommendation from the online return form

Urgent · 5 pt · backend · ml

> **As a** returns lead, **I want** a first-guess disposition on every online return, **so that** workers walk up to each item with a starting point, not a blank slate.

AC: • First-guess saved to return record within 30 s • Includes channel, confidence, expected $, and a short reason • Missing data lowers confidence rather than failing the call • Delta shown at inspection once condition is captured.

DIS-101 Capture condition at the warehouse & trigger the final call

High · 3 pt · frontend

> **As a** worker at the inspection station, **I want** to scan, grade, and see the routing recommendation on the same screen, **so that** I never re-enter anything between inspection and decision.

AC: • Confirming a condition grade fires the final recommendation call • Photo attach is optional and async — never blocks the recommendation • Failed scan falls back to typed item ID with confirm dialog • Walk-ins with no online return still work.

DIS-102 Live-data service · per-channel price, capacity, cost, demand

Urgent · 5 pt · backend · integration

> **As a** worker relying on the recommendation, **I want** every call to reflect the last 15 minutes of channel prices, capacity, and demand, **so that** the tool never routes toward a full or stale-priced channel.

AC: • Bundle returns < 200 ms p95, < 500 ms p99 • Missing pieces surfaced so the model can lower confidence rather than error • Demand data < 15 min old; older is flagged • Bundle shape versioned; breaking changes fail the build.

DIS-103 Generate the ranked recommendation with plain-English reason & confidence

Urgent · 8 pt · ★ MVP core

> **As a** worker at the moment of decision, **I want** the top recommendation on-screen within 400 ms of scanning with a reason I can trust, **so that** I confirm the call in flow instead of falling back to gut feel.

AC: • Ranked list of all 6 channels with reason, confidence, $ estimate; alternates pre-fetched for one-tap override • Top rec renders in < 400 ms p95 • Below 60% confidence, amber "ask a person" state with alternates expanded • Reason names ≥ 2 data points in plain English.

DIS-104 Confirm or override with a reason we can learn from

High · 3 pt · frontend · feedback-loop

> **As a** worker who knows this item better than the model does, **I want** to confirm with one tap or pick a different option and briefly say why, **so that** my expertise trains the next version without adding paperwork.

AC: • Top recommendation confirms with one tap or Enter • Overrides require selecting a reason (fixed list of 7) • Both original rec and new choice logged with the data snapshot • Reasons config-driven — a warehouse can add one without a deploy.

DIS-105 Persist every decision with its input snapshot

Medium · 3 pt · backend · analytics

> **As a** returns lead reviewing performance weekly, **I want** every decision — recommended, confirmed, overridden — in analytics within 24 h, **so that** reviews and retrains run on trustworthy data from day one.

AC: • Each decision stored with rec, confirmed action, reason, and input snapshot • Lands in the analytics warehouse within 24 h • Breakdown by warehouse, category, condition, override reason from day one • Schema versioned; breaking changes fail downstream builds.